

Lung and Colon Cancer Detection using Custom CNN

Abhinav Singh
EE25MT010

Neural Networks and Deep Learning
Indian Institute of Technology Dharwad, India

April 20, 2026

Outline

- 1 Introduction
- 2 Dataset
- 3 System Workflow
- 4 Model Architecture
- 5 Training
- 6 Results
- 7 Discussion
- 8 Conclusion

Problem Statement

- **Dataset:** Lung and Colon Cancer Histopathological Images.
- Contains 25,000 histopathological images of lung and colon tissues 5 classes:
 - Lung benign, Lung adenocarcinoma, Lung squamous cell carcinoma,
 - Colon adenocarcinoma, Colon benign.
- **Goal:** Classify images into these 5 cancer subtypes.
- Implemented a Convolutional Neural Network (CNN) from scratch.

Dataset Description

- **Source:** Lung and Colon Cancer Histopathological Images.
- **Total images:** 25,000 histopathological images.
- **Classes (5):**
 - Lung benign
 - Lung adenocarcinoma
 - Lung squamous cell carcinoma
 - Colon adenocarcinoma
 - Colon benign
- **Original image size:** 256×256 pixels (RGB).

Preprocessing and Augmentation

- Data split into 3 folders: Training, Validation, Testing.

Training Transforms:

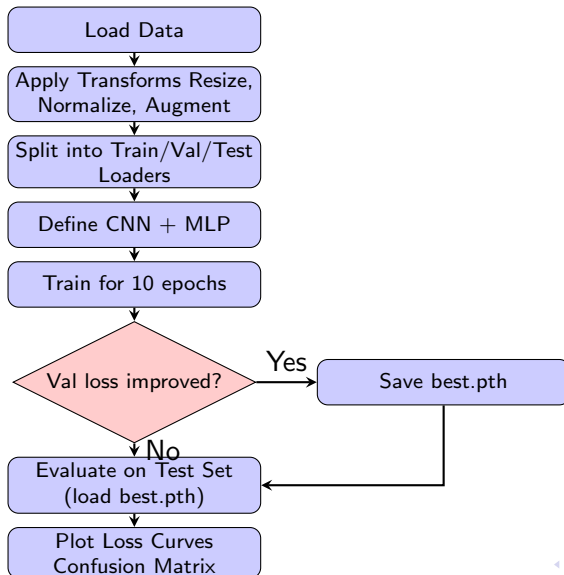
- Resize to 128×128
- Random Horizontal Flip
- Random Rotation(10°)
- To Tensor
- Normalize(mean=0.5, std=0.5) per channel

Test/Validation Transforms:

- Resize to 128×128
- To Tensor
- Normalize(mean=0.5, std=0.5) per channel

Note: RGB images (3 channels) – no grayscale conversion.

Complete Pipeline Flowchart



Convolutional Base

Layer	Output Size
Conv2d(3→16, kernel=3, padding=1)	$128 \times 128 \times 16$
BatchNorm2d + ReLU + MaxPool(2)	$64 \times 64 \times 16$
Conv2d(16→32, kernel=3, padding=1)	$64 \times 64 \times 32$
BatchNorm2d + ReLU + MaxPool(2)	$32 \times 32 \times 32$
Conv2d(32→64, kernel=3, padding=1)	$32 \times 32 \times 64$
BatchNorm2d + ReLU + MaxPool(2)	$16 \times 16 \times 64$

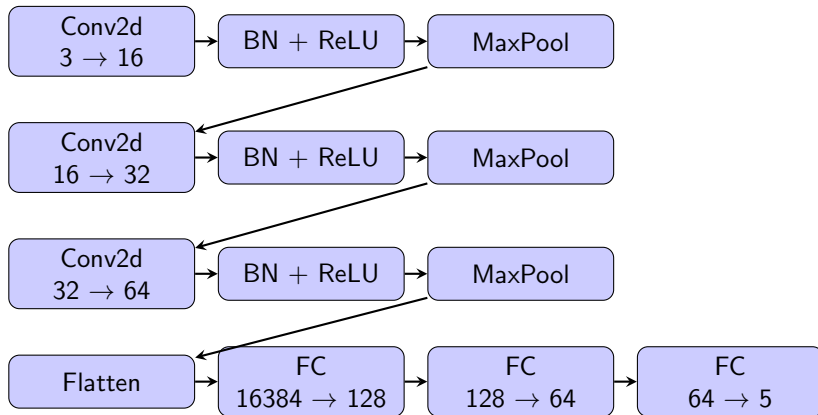
Output of CNN: $16 \times 16 \times 64$ feature maps.

MLP Classifier (Fully Connected Layer)

Layer	Output Size
Flatten	16384
Linear + ReLU + Dropout(0.3)	128
Linear + ReLU + Dropout(0.3)	64
Linear	5

Total trainable parameters: $\approx$ 2.1 million

Visualisation of CNN Blocks



Hyperparameters and Setup

- **Loss function:** CrossEntropyLoss
- **Optimizer:** Adam ($lr = 10^{-4}$, weight decay = 10^{-4})
- **Batch size:** 64
- **Epochs:** 10
- **Device:** MPS (Apple Metal)
- **Model checkpoint:** Saved when validation loss improves (best.pth)

Training and Validation Loss

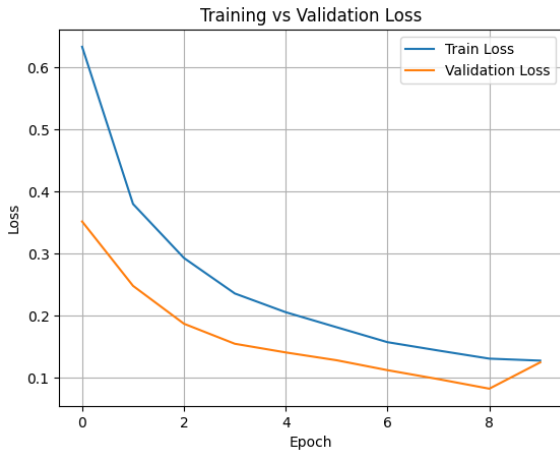
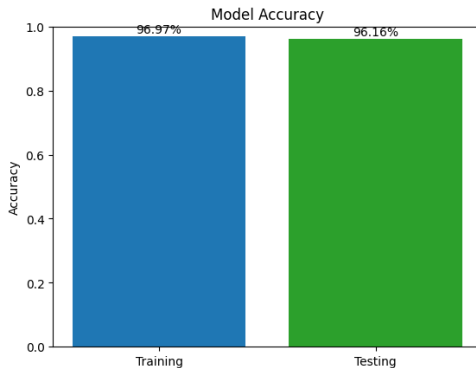


Figure: Loss vs. Epoch – both losses decrease steadily.

Accuracy on Train and Test Sets

Dataset	Accuracy (%)
Training	96.97%
Testing	96.16%



Confusion Matrix on Test Set

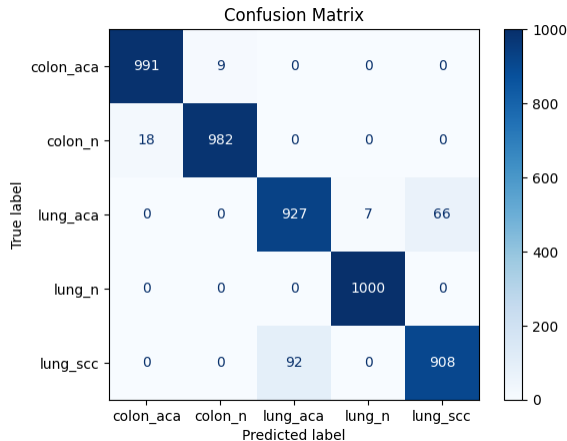


Figure: Per-class performance across 5 cancer subtypes.

- Training and validation losses decrease over 10 epochs.
- Using RGB images preserves color information, which may help distinguish tissues.
- Dropout (0.3) and weight decay (10^{-4}) effectively reduce overfitting.
- The best model is saved and loaded for final evaluation.

- Increase number of epochs or use early stopping.
- Apply stronger data augmentation (color jitter, elastic deformations).
- Use transfer learning with pre-trained models (ResNet, EfficientNet).
- Evaluate on external datasets for robustness.

- Designed and trained a custom CNN for 5-class histopathological image classification.
- Achieved testing accuracy 96.16%.
- Demonstrated end-to-end pipeline: data loading, augmentation, training, evaluation, and visualisation.

References



Andrew M. (2020).

Lung and Colon Cancer Histopathological Images.

Kaggle.

[https://www.kaggle.com/datasets/andrewmvd/
lung-and-colon-cancer-histopathological-images](https://www.kaggle.com/datasets/andrewmvd/lung-and-colon-cancer-histopathological-images)



Paszke, A. et al. (2019).

PyTorch: An Imperative Style, High-Performance Deep Learning Library.

In *Advances in Neural Information Processing Systems* 32.



Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012).

ImageNet Classification with Deep Convolutional Neural Networks.

In *Advances in Neural Information Processing Systems* 25.



Litjens, G. et al. (2017).

A Survey on Deep Learning in Medical Image Analysis.

Medical Image Analysis, 42, 60–88.

Thank You!